

An exact reduction from acyclic agreement forests to directed feedback vertex set

Reproducible rule reconstruction — awaiting expert review
22 September 2026

Abstract

We give deterministic polynomial-time maps from minimum-component acyclic agreement forests of two rooted binary phylogenetic trees to minimum directed feedback vertex set. Every globally minimum target solution recovers a globally minimum source forest, including all ties. Monotone ranks on tree vertices encode component membership and ancestral order. Parallel feasibility thresholds become unweighted bidirected cover graphs; a selector bounds the cover penalty at every infeasible threshold to one vertex. A directly validated feasible forest removes unneeded thresholds. For m augmented labels, the target has $O(m^4)$ vertices and arcs and $O(m^4 \log m)$ encoding bits. The contribution is an explicit, checked rule and decoder. No new complexity classification, priority claim, or minimum-overhead result is asserted.

1. Introduction

Acyclic agreement forests express how two rooted phylogenetic trees can be partitioned into compatible components without conflicting ancestral order. They connect the comparison of evolutionary trees to hybridization networks [1, §2]. An exact reduction to a graph optimization problem must preserve more than the existence of a feasible forest: an arbitrary optimal graph output must suffice to recover an optimal forest.

The closest structural reductions to directed feedback vertex set optimize splittings of a chosen agreement forest. The approximation guarantee in [1, §4] retains a factor-six comparison with the unrestricted optimum. Solving that restricted target exactly does not remove this loss. Polynomial SAT encodings for agreement forests and hybridization thresholds already exist [2, §§3–4]. SAT-to-cover constructions and bidirection are classical reduction tools [3].

This paper supplies a complete instance map and decoder for the unrestricted exact optimization endpoints. Its rank encoding replaces separate component membership and ancestral-order tables. Feasible upper bounds prune the threshold range, and a guarded formula bounds the cost of each infeasible threshold. These choices reduce the explicit graph compared with the component-slot baseline measured in the appendix. The construction does not establish a practical advantage over direct forest solvers or the best published encodings.

Theorem 1 (exact recovery). Let x consist of two rooted binary phylogenetic trees on the same nonempty leaf set, with an additional root label, and let m be the number of augmented labels. There are deterministic polynomial-time maps F and G such that $F(x)$ is an explicit unweighted digraph and, for every globally minimum directed feedback vertex set Y of $F(x)$, $G(x, Y)$ is an acyclic agreement forest with the minimum number of components. The graph has $O(m^4)$ vertices and arcs and $O(m^4 \log m)$ encoding bits. Neither map calls an optimization solver.

The construction encodes every relevant source threshold in a separate graph. A minimum solution of their disjoint union is minimum on each threshold graph. The graph at the true source optimum therefore supplies an optimal forest. Direct validation lets the decoder discard proposals from infeasible thresholds.

2. Definitions and output semantics

Let X be the original leaf set and $\rho \notin X$ the additional label. Augment each input tree by placing its root and a new leaf labelled ρ beneath a new root. Write $X^+ = X \cup \{\rho\}$ and $m = |X^+|$. Internal vertices of the two trees remain distinct, regardless of their input names.

For a nonempty block $B \subseteq X^+$, its connecting subtree is the minimal subtree joining its leaves, rooted at their lowest common ancestor. Its restriction suppresses vertices with one retained child. An agreement forest is a partition $\mathcal{P}$ of X^+ such that corresponding restrictions agree and the connecting subtrees are pairwise vertex-disjoint within each input tree. Its component ancestry graph has an arc $B \rightarrow C$ whenever the root of B is a strict ancestor of the root of C in either tree. The forest is acyclic when this union of ancestry relations is acyclic. Define $\text{OPT}(x)$ as the least number of blocks in such a forest.

A directed feedback vertex set of a digraph is a set whose deletion leaves an acyclic digraph. The target output must have globally minimum cardinality. The source output must have globally minimum component count. Both output sets are nonempty: the singleton partition is a source witness, and all vertices form a target witness. The empty set is the unique minimum target output on an empty graph. No no-solution output is admitted.

3. Ranks encode acyclic agreement forests

Fix an integer threshold k with $1 \leq k \leq m$. Assign every augmented tree vertex a rank $r(v)$ in $\{0, \dots, k-1\}$; copies of a labelled leaf share its rank. Impose three conditions. Ranks do not decrease along parent-to-child edges. Equal-rank leaves have an LCA of that same rank in each tree. Finally, three leaves cannot all have equal rank if their rooted triples differ between the two trees.

Lemma 2 (rank characterization). These conditions hold for some rank assignment if and only if an acyclic agreement forest has at most k blocks. Every such assignment yields a forest by grouping equal-rank leaves.

Proof. For a block B with at least two leaves, choose one leaf from each child branch of its connecting-subtree root. Their LCA is that root, so the LCA condition gives it the common leaf rank. Edge monotonicity forces this rank throughout all paths from that root to leaves of B . A singleton subtree is its leaf. Thus each connecting subtree is monochromatic. Different blocks have different ranks and cannot intersect.

Rooted triples determine a rooted binary labelled tree. To see this, a proper label subset C of size at least two is a cluster exactly when every pair $a, b \in C$ and every c outside C induce $ab \mid c$. The forward implication follows from the cluster root. For the converse, if C is not a cluster, its LCA has a descendant c outside C , while C meets both child branches. Taking a, b from those branches contradicts the triple condition. Triples therefore determine the cluster hierarchy. Restrictions to one or two leaves are unique. Excluding monochromatic disagreeing triples consequently makes the two restrictions of every block equal.

If a block root is ancestral to another, edge monotonicity gives a nondecreasing rank between them. Distinct blocks have distinct ranks, so every component ancestry arc strictly increases rank. The forest is acyclic and uses at most k blocks.

Conversely, take a forest with $q \leq k$ blocks. Number the blocks from 0 through $q-1$ in a topological order of their component ancestry graph, and assign these numbers to their leaves. At every other tree vertex, use the minimum rank of a descendant labelled leaf. This assignment is nondecreasing along edges.

Suppose v belongs to the connecting subtree of block B . A leaf of B lies below v . For a leaf below v from another block C , both component roots are ancestors of that leaf. If the root of C were at or above the root of B , the connecting path for C would intersect the subtree of B at v . Disjointness excludes this. The root of C is therefore strictly below the root of B , so C has larger rank in the topological order. The minimum descendant rank at v is exactly the rank of B . Every connecting subtree is monochromatic, which proves the LCA condition. Agreement of restrictions excludes disagreeing monochromatic triples. This gives all three rank conditions. $\square$

3.1. An explicit Boolean encoding

Use $w = \lceil \log_2 k \rceil$ bits for each rank, most significant bit first. Allocate leaf bits first, sharing them between the trees; allocate fresh bits for internal vertices. Bound each leaf rank by $k-1$. An internal

rank is bounded by every descendant leaf through edge monotonicity, so it needs no separate upper bound. For $k = 1$, bit vectors are empty and represent zero.

Equality is the conjunction of bit equalities. To compare unsigned vectors a and b , start with a true lower-bit comparison and process bits from least to most significant. At each position replace that comparison by the majority of the negated bit of a , the bit of b , and the previous comparison. Equal bits preserve the previous result; unequal bits set the comparison correctly. This proves the comparator by induction.

Encode every gate by both implication directions. Bit equality has its four truth-table clauses; majority has the six clauses saying that each true input pair forces the output and that the output requires a true input in every pair. OR gates and their dual AND gates use their usual equivalence clauses. A constant variable is fixed true, and a false clause is represented by its negative unit. Constant and repeated-literal simplifications preserve equivalence.

Assert the parent-child comparisons. For a leaf pair a, b with equality flag E_{ab} , assert

$$E_{ab} \implies r(a) \leq r(\text{lca}(a, b)). \quad (1)$$

The reverse inequality follows from tree edges, giving the LCA condition. Comparisons for a shared LCA and leaf are reused within a tree. For a disagreeing triple a, b, c , assert $\neg E_{ab} \vee \neg E_{ac}$. Call the complete CNF Φ_k . Lemma 2 and the exact gate equivalences prove that Φ_k is satisfiable exactly when a forest with at most k blocks exists. Every satisfying assignment yields such a forest.

4. The forward map

4.1. A certified threshold range

First compare the full restrictions. If they agree, return the empty graph; the source optimum is one. Otherwise the source optimum is at least two. Starting from singleton blocks, try all pairs of current blocks in deterministic order. Accept the first merge whose resulting partition passes the direct forest predicate, then restart. Stop when no merge is accepted. Each acceptance reduces the component count, so at most $m - 1$ merges occur. Let U be the final count.

Every accepted partition is feasible. Initially any two singleton labels can merge: their two-leaf restrictions agree, their connecting subtree meets no other labelled leaf, and every remaining component is a singleton with no outgoing ancestry arc. Consequently

$$2 \leq \text{OPT}(x) \leq U \leq m - 1. \quad (2)$$

Only thresholds $2, \dots, U$ are needed. The greedy procedure need not find an optimal forest.

4.2. Guarded formulas and cover graphs

For each $k < U$, add a fresh selector z to every clause of Φ_k and append the unit clause $\neg z$. Call the resulting CNF Ψ_k . It is equisatisfiable with Φ_k , and its satisfying assignments give the same leaf ranks. At $k = U$, use Φ_U unchanged because the feasible forest certifies satisfiability.

For any resulting CNF Θ , create two endpoints p_0, p_1 and an edge for each Boolean variable p . For each clause C of width d_C , create a private clique of occurrence vertices. Join every occurrence to the variable endpoint that makes its literal true. Let t be the number of variables and define

$$L_\Theta = t + \sum_{C \in \Theta} (d_C - 1). \quad (3)$$

All clauses are nonempty. Replace each undirected edge by its two opposing arcs. The resulting threshold digraph is H_k .

Lemma 3 (cover baseline and unit gap). Every cover has size at least L_Θ . If Θ is satisfiable, every minimum cover has size L_Θ and decodes to a satisfying assignment. A guarded formula has minimum cover size L_Θ when satisfiable and $L_\Theta + 1$ otherwise.

Proof. A cover selects at least one endpoint of each variable edge and at least $d_C - 1$ vertices in each private clique. These disjoint sets give Equation 3. A satisfying assignment attains it by selecting its truth endpoints and omitting one true occurrence per clause. Conversely, equality forces exactly one endpoint per variable and exactly one omitted occurrence per clause. The omitted occurrence's incident edge forces the corresponding true endpoint. Every clause is satisfied.

For a guarded formula, select both endpoints of z and one endpoint of every other variable. In each guarded clause omit the z occurrence and select all other occurrences; omit the negative unit's occurrence. All edges are covered, including the unit's edge to z_0 . This cover has size $L_\Theta + 1$. An unsatisfiable formula cannot attain the baseline, which proves the unit gap. $\square$

Figure 1 shows the complete graph for a contradictory one-variable formula after guarding. Selecting both selector endpoints supplies the single extra vertex permitted by Lemma 3.

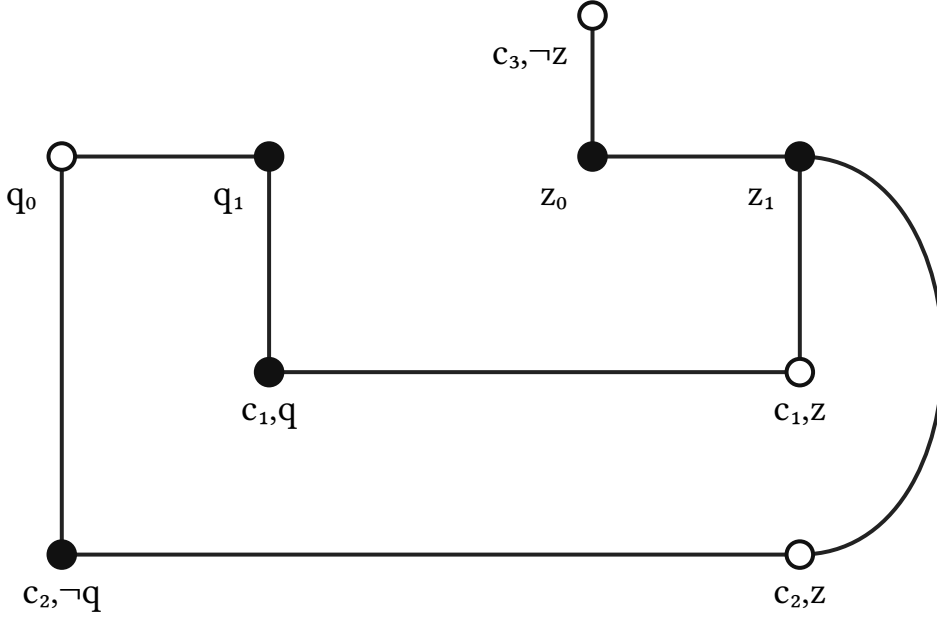


Figure 1: Exact cover graph for $(q \vee z) \wedge (\neg q \vee z) \wedge (\neg z)$. Endpoints encode variable values; $c_{j,\ell}$ is the occurrence of literal ℓ in clause j . Each line represents two opposing arcs. Filled vertices form a minimum cover of size 5; the baseline is 4. All 9 vertices and 9 undirected edges are drawn. The curved edge attaches the second z occurrence directly to z_1 . No edge is omitted.

A DFVS in a bidirected graph must hit every directed two-cycle, so it is a vertex cover. A cover leaves no arc and is a DFVS. Their feasible sets and minimum solutions are identical. Define $F(x)$ as the disjoint union of $H_2, \dots, H_U$, using distinct vertex namespaces. Figure 2 illustrates this composition.

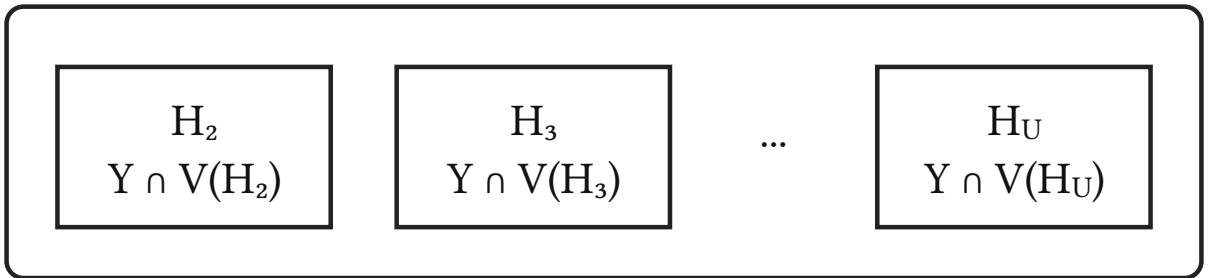


Figure 2: Schematic threshold composition for $U \geq 4$. Each box is the entire digraph H_k ; all internal vertices and arcs are omitted. There are no arcs between boxes. A global minimum DFVS Y restricts to a minimum DFVS in each box, since replacing a nonminimum restriction would improve Y .

5. Recovery from every minimum target solution

Given x and a minimum DFVS Y of $F(x)$, compare the full tree restrictions. If they agree, output the full-label block. Otherwise, for every $k = 2, \dots, m - 1$, read each leaf bit from membership of its true-value endpoint in Y . The numbering is reconstructed from the sorted label order and the bit width; selector and gate variables were allocated later. Group equal bit vectors into blocks.

Validate each proposal directly: it must partition the augmented labels, have equal restrictions, have vertex-disjoint connecting subtrees in each tree, and have an acyclic component ancestry graph. Among accepted proposals, return one with the fewest blocks, breaking ties by increasing threshold. Missing namespaces give all-zero ranks and propose the full-label block, which fails agreement on this branch. The decoder does not reconstruct the graph or rerun the greedy search.

Proof of Theorem 1. The forward construction gives an explicit legal graph. A global minimum DFVS restricts to a minimum DFVS on each threshold graph, by the disjointness shown in Figure 2. By Equation 2, the threshold $k = \text{OPT}(x)$ is present. Its formula is satisfiable, so Lemma 3 implies that every minimum target restriction decodes to a satisfying rank assignment. Lemma 2 gives a feasible forest with at most $\text{OPT}(x)$ components, hence exactly that many. There is therefore at least one accepted optimal proposal. Every other accepted proposal is feasible and cannot have fewer components. This includes proposals from infeasible thresholds, which need not encode satisfying assignments. The decoder returns an optimal forest for every minimum target output. The equal-tree branch handles source optimum one and empty target output. Polynomial bounds for both maps follow below. $\square$

6. Size and running time

At one threshold there are $O(m \log m)$ rank bits, $O(m^2 \log m)$ equality and comparison gates, and $O(m^3)$ disagreeing-triple clauses. The only gate of unbounded fan-in conjoins bit equalities. Before guarding, clause width is at most $\max(4, \lceil \log_2 m \rceil + 1)$; guarding adds one literal per original clause. The literal count is $O(m^3 + m^2 \log m)$, and the sum of squared clause widths is $O(m^3 + m^2 \log^2 m)$.

For t variables and clause widths d_C , the exact graph counts are

$$\begin{aligned} |V(H_k)| &= 2t + \sum_C d_C, \\ |A(H_k)| &= 2t + \sum_C d_C \cdot (d_C - 1) + 2 \sum_C d_C. \end{aligned} \tag{4}$$

There are at most m thresholds. Since $\log^2 m = O(m)$, their union has $O(m^4)$ vertices and arcs. Generated indices use $O(\log m)$ bits, giving $O(m^4 \log m)$ explicit encoding length. No weight is expanded or stored implicitly.

The greedy upper-bound search checks $O(m^3)$ proposals. Each direct forest validation takes at most $O(m^3)$ combinatorial operations, with polynomial bit and string costs. In the implemented path representation, a pair-LCA query scans $O(m)$ ancestors and performs tuple membership tests of $O(m)$ cost. All pair queries at a threshold take $O(m^4)$ work; all triple queries take $O(m^5)$. Across all thresholds, $O(m^6 \text{ polylog}(m))$ is a conservative bound for the forward combinatorial and index work. Parsing and sorting arbitrary input strings add polynomial work in the actual source encoding length.

The decoder reads the target output into a set, reconstructs leaf-bit indices, and validates fewer than m partitions. The conservative bound is $O(m^5)$ combinatorial work plus polynomial bit and string work in the combined input lengths. The output lists each source label once. Tree traversal is iterative. Restrictions use delimited leaf indices and sorted, parenthesized child strings, with unary vertices suppressed, so tree depth does not invoke a fixed call-stack limit. Both maps are deterministic and polynomial in their required input sizes.

7. Scope and limitations

The rule supplies exact recovery for unrestricted source forests and all target optima. Its bidirected targets are vertex-cover instances. It establishes no approximation guarantee, parameterized improvement, optimal size bound, or practical speed superiority. Polynomial SAT encodings and generic reduction composition already imply broad existence results; the contribution here is the explicit rank construction, decoder, proof, and checked implementation. Priority for the rank characterization or selector technique remains unassessed.

References

- [1] Steven Kelk, Leo van Iersel, Nela Lekić, Simone Linz, Celine Scornavacca, and Leen Stougie. *Cycle killer... qu'est-ce que c'est? On the comparative approximability of hybridization number and directed feedback vertex set*. arXiv:1112.5359v1, 2011. Primary manuscript.
- [2] Maria Luisa Bonet and Katherine St. John. *Efficiently Calculating Evolutionary Tree Measures Using SAT*. Author-hosted manuscript, §§3–4, printed pp. 6–8. Year and venue are not specified here because the consulted copy did not establish them. Primary manuscript.
- [3] Richard M. Karp. *Reducibility among Combinatorial Problems*. In *Complexity of Computer Computations*, pp. 85–103, Plenum Press, 1972. doi:10.1007/978-1-4684-2001-2_9.

A. Verification and reproducibility

A.1. Evidence and finite limits

The tested implementation is `work/algorithm.py` at Git revision 25cab27. The proof’s local LCA accounting was corrected in 5f3a30b without changing either map. The independent review and focused follow-up are retained in `reviews/rank-threshold/`. The review advances this rule for expert assessment; it is not a formal proof certificate or publication acceptance.

The prepared family has 327 input records: all 236 ordered labelled binary-tree pairs through four original leaves, 80 seeded pairs on five through seven leaves, nine historical inputs, and two explicit regressions. Source oracles independently agree on 113,194 partition predicates and all 956 minimum partitions. A separate exhaustive check agrees with all 766 minimum outputs on 631 small directed graphs. The larger tree pairs use seed 20260922; the committed input file fixes the cases.

The complete current candidate loop executes both maps as subprocesses and independently solves each emitted target. All 327 inputs and 38,427 minimum-target recoveries passed. The 300 nonempty-target records stop at 128 outputs each; the 27 empty targets are exhaustively checked. Thus these counts do not represent all target ties. Eight disjoint shards cover the exact case-name set and all exit successfully. A further independent CP-SAT implementation passes 12 input records and 33 recoveries. Both maps also pass identical-tree regressions at 1, 2, and 1,200 leaves with renamed nodes and reversed child orders.

The reviewer performed 47 targeted recovery calls on three- and four-leaf pairs. These exhaust the selected optimal-threshold rank projections and include all 32 binary leaf-rank patterns at an infeasible threshold. They recover all source optima on those pairs. This projection coverage is distinct from complete DFVS-set enumeration. Solver unknowns and interrupted runs are never counted as optimum answers. All tests have finite instance or output bounds; none uses a solver or process timeout. The universal statement rests on the proof.

A.2. Measured graph overhead

Table 1 compares the current explicit graphs with the earlier component-slot baseline on the same stored inputs. Each cell is a median over the indicated number of records. The iterative traversal repair preserves every vertex and arc on this family. The improvement is relative to this implementation baseline; no lower bound on possible overhead or comparison to the best published encoding is claimed.

$ X $	Records	Baseline vertices	Current vertices	Baseline arcs	Current arcs
4	231	12,664	546	39,698	1,610
5	33	25,889	3,081	82,522	10,796
6	32	46,790.5	7,452	151,156	28,930
7	16	77,799	15,267.5	254,150	60,440

Table 1: Graph-size medians on fixed records. Source counts include regression records that overlap the exhaustive topological family. The current worst seven-leaf graph has 23,437 vertices and 96,142 arcs.

These remain large expansions of small source trees. Timing measurements in the retained comparison belong to the previous recursive implementation and are not benchmarks of the current maps or end-to-end solver performance.

A.3. Environment and commands

From the repository root, use Python 3.12.11 and uv 0.12.7. The committed lockfile pins Z3 4.15.4.0, OR-Tools 9.15.6755, and NetworkX 3.6.1. The maps themselves use only the Python standard library. Typst 0.15.1 compiles this paper. Lean and Lake are available, but no Lean formalization is claimed.

```
uv sync --locked --python 3.12.11
cd campaigns/acyclic-agreement-forest-feedback-vertex-set/work
uv run --locked python check.py --self-test
uv run --locked python check.py --candidate algorithm.py
uv run --locked python verify.py --candidate algorithm.py
uv run --locked python ../rounds/006/deep_tree_check.py
uv run --locked python ../rounds/006/gadget_check.py
uv run --locked python ../rounds/006/reproduce_modes.py
```

The recorded scheduling appends `--shard I 8` to the candidate-check command for each integer I from 0 through 7. The unsharded command checks the same family. The minimal reproduction script selects a stored three-leaf input, executes the forward command, obtains a minimum target solution independently, and executes extraction. Its target has 352 vertices and 1,024 arcs; its minimum DFVS size is 193, and recovery returns an optimal two-component forest. It tests one minimum target output and does not exhaust that target’s optima.

For a legal source JSON document, the two public command modes are:

```
uv run --locked python algorithm.py < source.json > target.json
uv run --locked python algorithm.py --extract \
    < recovery.json > forest.json
```

The extraction input `recovery.json` contains the original source object and a `target_solution` object. The latter has `problem` equal to `dfvs` and `feedback_vertex_set` equal to a globally minimum deletion list for the actual `target.json`. The reproduction script constructs this payload automatically. `contract.md` specifies all JSON fields and the exact legal domain.

The proof, preparation report, verification report, and Round 006 logs retain full evidence and limitations. The independent review includes primary-source locations and its bounded literature coverage. Its checks used a fresh reviewer context with instruction-based write confinement; no filesystem sandbox or separate model was claimed as a correctness guarantee.

```
uv run --locked python figures/draw_gadgets.py
typst compile manuscript.typ manuscript.pdf
```

The figure generator checks the exact gadget’s adjacency and minimum cover by exhaustive subsets. The compiled PDF is inspected page by page for mathematical notation, figure legibility, clipping, and layout. These presentation checks do not extend the theorem’s formal verification status.